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6 SHORT TITTLE:

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9 AUTHORS:

10 1. Eduardo de Faria Castro Fleury, MD, PhD. Professor Centro Universitário São

11 Camilo, curso de Medicina – São Paulo/ Brazil;

12 2. Instituto Brasileiro de Controle do Câncer – São Paulo/ Brazil.

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14 CORRESPONDENCE:

15 Rua Maestro Chiaffarelli, 409 – Jardim Paulista – São Paulo/ Brasil. ZIP: 01432-030

16 Phone: +55 11 98148 11 88

17 Email: edufleury@hotmail.com

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34

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48 Silicone Induced Granuloma of Breast Implant Capsule (SIGBIC) diagnosis: Breast
49 Magnetic Resonance (BMR) ability to detect silicone bleeding.
50
51 ABSTRACT:
52 Objective: To evaluate the ability of BMRI to detect silicone gel bleeding in a
53 prospective observational study including consecutive patients referred for BMRI scan.
54 Methods: From January 2017 to March 2018, patients referred for BMRI were
55 evaluated in a prospective observational study. Patients who had breast implants were
56 included. BMRI recorded 9 findings according to BI-RADS lexicon and SIGBIC findings,
57 considered equivocal features to detect gel bleeding (GB). Three new original imaging
58 features were added for SIGBIC diagnosis: black drop signal; T2* hypersignal mass; and
59 delayed contrast enhancement, considered as irrevocable signs. The presence of
60 silicone corpuscle was confirmed by percutaneous biopsy or surgical capsulectomy.
61 Accuracy of BMRI SIGBIC findings to predict GB was determined. We also used
62 univariate analysis for the equivocal features for GB diagnosis. The Backward method
63 was applied for a multivariate Logistic Regression model for the equivocal features.
64 Results: SIGBIC was diagnosed in 208 patients and GB was histologically confirmed in
65 all cases. No false positive results were observed. The most important imaging
66 equivocal feature associated with GB was capsular contracture. In order of prevalence,
67 the main equivocal BMRI features associated to GB with statistically significance ($P <$
68 $=0.001$) were as follows: 1.water droplets (OR=2.8; 95%CI 1.8-4.4); 2.enlarged
69 intramammary lymph node (OR=3.1; 95%CI 1.5-6.1); 3.pericapsular edema (OR=5.0;
70 95%CI 2.3-11.1); and 4.intracapsular seroma (OR=2.4; 95%CI 1.4-4.1).

71 Conclusion: SIGBIC diagnosis has high sensitivity to predict GB by the 3 irrevocable
72 BMRI features described by the authors. We suppose GB is underdiagnosed in clinical
73 practice by BI-RADS features.

74

75 **Keywords:** Breast implant, granuloma, silicone elastomers, Magnetic Resonance
76 Imaging, Anaplastic Large-Cell Lymphoma

77

78 **List of abbreviations:** SIGBIC, silicone-induced granuloma of breast implant capsule;
79 BMRI, breast magnetic resonance imaging; GB, gel bleeding; BIA-ALCL, breast implant-
80 associated anaplastic large cell lymphoma; EILN; enlarged intramammary lymph node.
81 SIIS; silicone implant incompatibility syndrome.

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95 INTRODUCTION

96 Over the last 2 years, an increasing number of studies have reported
97 complications related to breast silicone implants (1–4). These complications are
98 commonly associated with the incidence of breast implant-associated anaplastic large
99 cell lymphoma (BIA-ALCL) (5–12); however, the onset of this pathology and its trigger
100 point are yet to be elucidated.

101 Interestingly, many of these changes have been reported despite the
102 macroscopic integrity of breast implants. The main complications are reported as:
103 breast stiffness, late seroma, lymph node enlargement, and silicone migration to
104 distant organs. In addition, clinical manifestations related to autoimmune reactions
105 have been reported, which highlight the silicone implant incompatibility syndrome
106 (SIIS) (13).

107 However, gel bleeding is virtually ignored by the academy as a trigger point for
108 these changes. Most articles advocate that gel bleeding is a rare event with no
109 pathological significance. There are no studies in the literature demonstrating the
110 frequency of this event in BMRI scans.

111 Recently we have described a new radiological finding, silicone-induced
112 granuloma of breast implant capsule (SIGBIC) (14), which is defined as granulation
113 tissue formed from a reaction between the breast implant fibrous capsule and free
114 silicone corpuscle due to bleeding of the intact breast implant. We have described 3
115 BMRI features that are irrevocable for this diagnosis: 1. black-drop signal; 2. mass with
116 hypersignal at T2-weighted sequences; and 3. late contrast enhancement (15). In
117 addition, we described the mechanism of its intracapsular formation (16).

118 The aim of this study was to determine the ability of BMRI for predicting GB
119 according to SIGBIC diagnosis with the 3 irrevocable features. Furthermore, we
120 determined if the equivocal imaging features according to the latest BI-RADS lexicon
121 could predict GB. This is the first prospective study to determine its prevalence in
122 clinical practice.

123

124 **MATERIALS AND METHODS**

125 This was a prospective observational study conducted at a single institution
126 from March 2017 to August 2018. We evaluated patients referred for BMRI. All
127 patients who had breast implants were included in the study. The exclusion criteria
128 were patients with previous BMRI scan during the study period to avoid duplicity of
129 the results, technically inappropriate scans, and scans without contrast injection. The
130 study was conducted in accordance with local ethics committee regulations. Free
131 informed consent was obtained from all patients.

132 We used a standard BMRI protocol for a 1.5-T imager (Magnetom Aera;
133 Siemens Healthcare) with a dedicated eight-channel bilateral breast coil in the axial
134 orientation. The acquired sequences were: 1. Axial T2-weighted fast spin-echo; 2.
135 Sagittal Proton-density weighted; 3. Sagittal T2-weighted silicone selective that
136 enhance the silicone signal; 4. Sagittal T2-weighted sequence with silicone suppression
137 that extract the silicone signal; and 5. 4 sequential dynamic contrast sequences with
138 fat suppression and reconstruction with subtraction.

139 BMRI images was read by 1 radiologist with 18 years of experience in BMRI.
140 Nine BMRI findings were reported according to the BI-RADS lexicon, considered as
141 equivocal signs:

- 142 1. Capsular contracture: when the anteroposterior diameter of the implant is
143 increased with fibrous capsule thickening and contrast enhancement
144 (Figures 1, 2, 3, and 4).
- 145 2. Intracapsular rupture: discontinuity of the breast implant surface, often
146 characterized by: subcapsular line, keyhole sign, teardrop sign, and linguine
147 sign restricted to intracapsular space.
- 148 3. Extracapsular rupture: discontinuity of the fibrous capsule with
149 extravasation of the internal silicone contents.
- 150 4. Water droplets: foci of water signal inside the intact breast implant
151 inferring loss of surface permeability (Figure 1).
- 152 5. Implant Rotation: displacement of the posterior surface of the breast
153 implant, usually identified by the seal (Figure 2).
- 154 6. Extracapsular Siliconoma: extracapsular free silicone without signs of
155 fibrous capsule rupture.
- 156 7. Enlarged intramammary lymph node (EILN): enlarged pericapsular
157 intramammary lymph node (Figure 3). Two criteria were used to determine
158 EILN: smaller diameter greater than 0.5cm and thickening of the cortical
159 with reduced fatty hilum.
- 160 8. Pericapsular edema: non-mass pericapsular enhancement (Figure 4).
- 161 9. Intracapsular seroma: collection inside the fibrous capsule (Figures 1 and 2).

162 In addition to the features described by the BI-RADS lexicon, a further 3 new
163 original features were incorporated that are reported with the presence of SIGBIC,
164 considered as the tenth finding:

- 165 - 10.a. black drop signal: marked focus of hiposignal at T1 sequences in the
- 166 fibrous capsule, silicone signal focus could be associated at silicone
- 167 sensitive sequences, without enhancement in pos contrast sequences;
- 168 - 10.b. mass with hypersignal at T2 weighted sequence: intracapsular mass
- 169 that could misdiagnose as seroma
- 170 - 10.c. late contrast enhancement: the mass (10b) shows enhancement at the
- 171 late phase.

172 SIGBIC diagnosis was only determined when all the 3 irrevocable features were

173 present.

174 Patients diagnosed with SIGBIC at BMRI underwent a second-look ultrasound

175 scan using a device with 7.5–14 MHz multi-frequency probe (Aplio 300, Toshiba). All

176 the patients who meet the SIGBIC BMRI diagnostic criteria underwent US-guided

177 percutaneous core-biopsy or were directed for surgical capsulectomy. It was opted for

178 the percutaneous biopsy in patients who have easy access for performing the

179 procedures. In patients where the lesions were very deep or posterior to the breast

180 implants, capsulectomy was opted. Percutaneous biopsy was performed using a 14-G

181 needle attached to an automatic biopsy gun. At least 3 samples were collected. The

182 remaining BMRI findings were followed up.

183 The specimens from biopsies or surgical procedures were evaluated in the

184 same institution, where a diagnosis of SIGBIC was confirmed when the silicone particle

185 was observed at microscopy, as described in a previous study. (Figure 5)

186 Patients SIGBIC diagnosis were further divided into groups and compared to the

187 SIGBIC findings, as follows:

- 188 - Type of surgery: aesthetic or reparative;

189 - Type of implant: silicone, saline, or double lumen;

190 - Location of the implant: retroglandular or retropectoral;

191 - Implant placement time.

192 The main objective of the study was to determine the ability of BMRI to predict
193 gel bleeding in patients with SIGBIC diagnosis compared with histopathology as the
194 gold standard. As secondary objectives we established what equivocal BMRI features
195 could be related to GB. In this context, a univariate analysis was performed through
196 Chi-Square tests (17) and Fisher's Exact Test (18) and test of Mann-Whitney (19) for
197 the case of categorical variables (20). The approach used was not the automatic
198 Stepwise. Through the univariate analysis, the potential predictors for the response
199 variable were selected, being considered a level of significance equal to 25% according
200 to Kennedy and Bancroft criteria (21). The multivariate Logistic Regression model was
201 only used for patients with histopathological proved SIGBIC in order to determine
202 diagnosis ability using ONLY the established BI-RADS lexicon BMRI features.

203 We also evaluated, in order to verify if the adjusted models were adequate and
204 if they had good predictive capacity, some measures of quality of adjustment using
205 only the association of the main equivocal features for GB diagnosis: Pseudo R^2 (22),
206 AUC (area under the ROC curve), Sensitivity, Specificity, VPP, and the Hosmer-
207 Lemeshow test (18). The software used in the analyzes was R (version 3.5.2).

208 For illustrative purposes, a descriptive analysis was also performed of the
209 appropriate variables of interest. The absolute and relative frequencies were used.
210 Whereas, for numerical variables description, positions, central tendency and
211 dispersion were applied.

212

213 RESULTS

214 We performed 2290 consecutive BMRI from March 2017 to August 2018. Of
215 these, 736 patients had breast implants. Fifty-six patients were excluded from the
216 study, 48 because they refused the injection of contrast medium, 6 because the
217 presence of movement artifacts or difficulty in performing the silicone sequences, and
218 2 due to previous BMRI scans during the study period. We also excluded 472 patients
219 without the 3 diagnostic criteria for SIGBIC (equivocal features)

220 The remaining 208 patients with SIGBIC diagnosis by BMRI (irrevocable signs)
221 were included in the study. All 208 patients with the irrevocable findings had GB
222 confirmed by silicone corpuscles in histopathology. Histological specimens of 40
223 (19.9%) cases were obtained by surgical capsulectomy and 168 (80.1%) from
224 percutaneous breast biopsy. There were no false positive results of GB adopting the
225 SIGBIC irrevocable criteria. All capsular specimens showed silicone particles associated
226 with inflammatory response at histology, and all patients submitted to surgical
227 capsulectomy have no macroscopic signs of implant rupture.

228 Univariate analysis:

229 Table 1 shows the univariate analysis using Chi-Square test (17) for categorical
230 explanatory variables with expected frequencies greater than 5 in all class and Fisher's
231 exact test (17) for the opposite cases. For the numerical variables (patient age and
232 time of surgery), Mann-Whitney test analysis (20) was performed. The equivocal
233 features with statistical significance selected in the univariate analysis ($p < 0.001$) to
234 predict GB diagnosis were: water droplet, enlarged intramammary lymph node,
235 pericapsular edema, and intracapsular seroma. Type of surgery (p -value = 0.023),
236 Implant type (p -value = 0.066) and patient age (p -value = 0.026) were also selected,

once which had a p-value of less than 0.250. The variable capsular contracture did not participate in this analysis since it was observed in all patients.

Multivariable analysis:

A multivariate model of Logistic Regression (17) was adjusted from the variables selected in the univariate analysis and for this model the Backward method was applied for the final selection of the variables, considering a significance level of 5%.

Table 2 presents the final multivariate model for GB variable with the equivocal features described by the BI-RADS lexicon. When positive an individual has the chance to have GB multiplied by 2.8 [1.8; 4.4] in the presence of water droplet, 3.07 [1.5; 6.1] in the presence of enlarged intramammary lymph node, 5.02 [2.3; 11.1] in the presence of pericapsular edema, and 2.4 [1.4; 4.1] for intracapsular seroma. Examination indication was significant (p-value <0.001) too, and considering the screening category as a reference, an individual with a diagnostic indication has a chance of having GB multiplied by 5.1 [2.6; 10.0].

Table 3 shows the negative predictive value (NPV), Hosmer-Lemeshow test and Pseudo-R² test for determination of GB for the final model adopting the most statistic significant equivocal BMRI features. It should be noted that: the area under the ROC curve (AUC) of the final model was 0.750 and the Pseudo-R² was 24.79%. And the Hosmer-Lemeshow test for the final model indicated that the fit of the model was adequate (p-value = 0.795).

Descriptive analysis:

261 It is observed that all patients with SIGBIC features had silicone corpuscles at
262 histology and capsular contracture at BMRI. 18.7% had water droplet signs, 12.1% had
263 intracapsular seroma and 7.1 had enlarged intramammary lymph node. The diagnostic
264 indication for BMRI scan was prevalent (82.35%). The mean patient age value was 42.3
265 years with a standard deviation of 11.2 and the last surgery procedure was on average
266 7.4 years ago.

267 The mean time for patients diagnosed with SIGBIC was 7.7 years (median, 6
268 years). When evaluated if there is association between the diagnostic BMRI scan to the
269 years passed after implant placement, the association to SIGBIC was 12 years ($P =$
270 0.0346) after surgical procedure for all patients. When we split these patients and
271 assessed patients undergoing cosmetic surgery the association was 12 years ($P=0.052$)
272 while that who underwent oncologic procedure was 7 years ($P=0.0485$). However,
273 when evaluating implant location, we found no statistically significant difference in
274 presence of SIGBIC between retroglanular or retromuscular implants.

275

276 **DISCUSSION**

277 Currently, silicone bleeding is virtually ignored by academia as a possible trigger
278 factor for complications related to silicone implants. Most articles illustrate silicone
279 bleeding as rare event and of little clinical relevance. They justify this statement due to
280 the evolution of breast implants, especially with the use of cohesive gel.

281 Since the description of SIGBIC by our group we observed an increasing
282 prevalence of SIGBIC in our clinical practice; therefore, we conducted this prospective
283 study to determine BMRI ability to predict GB in patients with SIGBIC diagnosis. We

284 also intended to assess the association between BMRI equivocal features described by
285 the latest BMRI BI-RADS lexicon and SIGBIC in predicting GB.

286 Reported breast implant-associated complications have recently been
287 increasing, including BIA-ALCL and Breast Implant Illness in academia, mainstream and
288 social media. Despite the reported severity associated with anaplastic lymphoma, the
289 number of cases described in the literature remains small, with <650 cases reported,
290 minimizing its relevance (11,12). Furthermore, the trigger point to develop this
291 pathology remains unclear.

292 We recently described the radiological findings of a granuloma developed in
293 the intracapsular compartment of breast implants that was formed by an immune
294 response of the fibrous capsule to the free silicone particles bled from intact breast
295 implants (16). All breast implants after a certain time or when subjected to stress
296 situations (like heat and trauma) may alter the surface permeability may present gel
297 bleeding of the internal contents or leakage of the shell (1). Both the silicone and
298 saline implant have the substance polydimethylsiloxane (PDMS) as a shell component,
299 and it is speculated that this substance can elicit an immune response when in contact
300 with the fibrous capsule. Some patients have a higher risk for developing immune
301 response, for example, those with SIIS. SIIS clinical manifestations may vary and
302 depend on the intensity of the inflammatory reaction of the host. (16).

303 In our study, 30.6% patients with breast implants referred for BMRI scan at our
304 facility fulfill the irrevocable BMRI criteria for SIGBIC diagnosis. All these cases were
305 confirmed by histopathology by the presence of silicone corpuscles.

306 Due to the novelty of the study, we chose very restrictive criteria for the SIGBIC
307 diagnosis. False positive results were not desired in this context. The black-drop signal

308 consists of giant cell reaction to a foreign body in the fibrous capsule. The mass with
309 hypersignal at T2-weighted sequence is the development of granulation tissue in the
310 contacted area between the silicone corpuscle and the fibrous capsule. Finally, the late
311 contrast enhancement is because of the poor intracapsular vascularization due to the
312 fibrous capsule protective barrier and could differentiate the granuloma from
313 intracapsular seroma. When the three criteria were met, we could predict GB in all
314 cases.

315 We found statistical significance association between GB and some of the
316 equivocal BMRI BI-RADS lexicon features: capsular contracture, water droplets,
317 enlarged intramammary lymph node, pericapsular edema, and intracapsular seroma.
318 All probably due to leakage of silicone gel in intact implants. When used the Hosmer-
319 Lemeshow test and pseudo-R² to associate the presence of GB with only the equivocal
320 BMRI features, there was found good sensitivity, specificity and diagnostic accuracy for
321 predicting GB. These findings could support the hypothesis of GB be underdiagnosed in
322 our clinical practice where most of the articles states that is a rare event.

323 We hypothesized that water droplets are related to permeability loss in the
324 breast implant shell. When intracapsular fluid enters through the implant shell, a
325 chemical reaction occurs with the inside silicone content or the silicone-made surface
326 of the implant, which leads to silicone residues into the intracapsular compartment.
327 Water-droplet corresponds to a macroscopic diagnosis, where the findings are seen
328 with the naked eye by BMRI. It is speculated that the water-droplets diagnosis by BMRI
329 may be underestimated if compared with microscopy. (Figure 6) These findings were
330 similar to that described as: “*breast implants, from clear to cloudy*” (23) where the
331 authors report color changes of intact breast implants due to a chemical reaction with

332 intracapsular fluid. In our study, when water droplets are present the risk of SIGBIC is
333 increased in 2.82.

334 There was also a correlation between surgery type and GB. Oncologic surgery
335 time and SIGBIC diagnosis up to 7 years after surgery, which may be due to greater
336 manipulation of the implant in therapeutic surgery, increased surgical times, and
337 radiotherapy sessions (28).

338 We observed a high prevalence of GB in our study. Interestingly, no previous
339 study has reported this occurrence neither the relation of GB with BMRI findings.
340 There may be different reasons for this. First, culturally most BMRI protocols for breast
341 implant evaluation are performed without the use of contrast media. This factor may
342 lead to a misdiagnosis of SIGBIC as late seroma. Second, there may be a lack diagnostic
343 expertise. For example, before 2017, our service had not reported a positive diagnosis
344 of SIGBIC. However, some patients were diagnosed with SIGBIC during the study
345 period, following persistent clinical complaints, and when comparing to previous
346 exams; SIGBIC was retrospectively diagnosed.

347 The relevance of this manuscript is the novelty of the issue and to alert to the
348 possibility of gel bleeding as a precursor to complications inherent to silicone breast
349 implants. The results exhibit the high frequency of gel bleeding in this study, where it
350 describes the restricted criteria for diagnosis. Currently, many diseases are related to
351 silicone implants, such as BIA-ALCL, SIIS, and Breast Implant Illness (BII), without a clear
352 hypothesis of what is the trigger point for its development. With the information
353 obtained from the BMRI, the doctor can more accurately manage patients with
354 complaints related to breast implants.

355 This study has limitations. First, we only evaluated diagnostic BMRI in a single
356 center. Future prospective multicenter studies should be conducted to confirm our
357 findings. We are initiating a multicenter study involving three different breast
358 diagnostic centers to validate our results. An additional limiting factor was that only
359 patients diagnosed with SIGBIC underwent percutaneous biopsies or surgical
360 capsulectomies. As an observational study, it was not accepted by our ethical
361 committee to have a control group submitted to breast biopsy. However, we were able
362 to illustrate that MRI can diagnose gel bleeding, the novelty of this study. Future
363 studies should access the histology of patients with capsular contracture to search for
364 free silicone in the fibrous capsule. It is important to emphasize the importance of
365 pathologist and radiologist training to diagnose SIGBIC.

366 Our study supports that GB prediction could be performed adopting the 3
367 irrevocable SIGBIC criteria proposed by the authors. It also demonstrates that
368 equivocal BMRI features related to GB were capsular contracture, water droplets,
369 enlarged intramammary lymph node, pericapsular edema and intracapsular seroma.
370 Based on our findings, we suppose it is underdiagnosed in clinical practice and could
371 explain most of the implant reported complications.

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450

451 **Figure legends**

452

453 Figure 1. A 45-year-old woman 6 years after implant placement with left breast

454 stiffness

455 T2-weighted and silicone-sensitive images on the right and left, respectively. Blue and

456 red arrows demonstrate an intracapsular seroma and mass that includes free silicone.

457 (a) Intact breast implant. (b) Green arrow demonstrates water droplets.

458

459 Figure 2. A 30-year-old woman who underwent an aesthetic procedure for silicone

460 prosthesis four years ago, 1 week before an increase in volume and inflammation of

461 the right breast

462 (a) STIR sequencing presented a massive intracapsular collection (red asterisk) with

463 intact and rotated breast implant (green asterisk), and free silicone in the fibrous

464 capsule forming the granuloma (yellow arrow). (b) Identical results were found with

465 T2W silicone-sensitive, and after the use of contrast. (c) Findings were compatible with

466 the black drop signal. Intact breast implant.

467

468 Figure 3. A 62-year-old patient underwent right breast repair surgery three years ago,

469 followed by treatment with radiotherapy sessions

470 (a) Arthralgia had been reported for 3 months. Intracapsular masses are shown in the

471 pre-contrast sequence (green arrow), with enlarged intramammary lymph nodes (red

472 arrow), and black-drop signal (yellow arrow). (b) A left mass was revealed by

473 hypersignal on T2 imaging. (B). Intact breast implant.

474

475 Figure 4. (a, b, c, d, e and f) A 37-year-old patient with a breast implant for 3 years,
 476 who showed a palpable mass in the medial quadrants of the left breast for 3 weeks
 477 (a) T2W fat-suppressed imaging showing a capsular contracture with a hypersignal
 478 mass in the medial quadrant and pericapsular edema (yellow arrow). The post-contrast
 479 sequence shows the late enhancement of this mass (green curve) compared with the
 480 heart (blue curve). (b) A pericapsular enhancement is shown (yellow arrow). (c)
 481 Ultrasound with elastography shows siliconoma at the site *via* BMRI, characterized by
 482 a hard mass at and snowstorm artifacts (yellow arrow), with a small collection at the
 483 periphery (green arrow). (d) Mammography shows a pericapsular mass in the medial
 484 quadrant (yellow arrow). (e) Histology showing free silicone particles and inflammatory
 485 cells (magnification, 50×). Immunohistochemistry for T lymphocyte (CD3) positive.
 486 Intact breast implant.

487 Figure 5. Examples of percutaneous breast biopsy in 2 patients. Patient 1 (A, B, C and
 488 B) and Patient 2 (e, f, g and h). At BMRI (A and E), findings suggestive of SIGBIC pointed
 489 by the yellow arrow. Ultrasound elastography (B and F) showing a hard mass at the
 490 fibrous capsule (green arrow). Biopsy of the mass (C and G) using first a fine needle
 491 aspiration to collect the intracapsular seroma (C) and core biopsy of the mass (G)
 492 represented by the white arrow. Histological specimens confirming silicone granuloma
 493 in red arrow.

494 Figure 6. Correlation between water droplets at BMRI and microscopy of implant shell.
 495 At BMRI (a) is possible to see a focal signal change inside the breast implant pointed by
 496 the blue arrow. At the microscopy evaluation of implant shell (b) is possible to note
 497 silicone leakage pointed by the red arrow. Inside the implant with a 1.600-fold
 498 microscopy (c) it is possible to see the water droplets formed from a chemical reaction

499 between the intracapsular fluid and the silicone gel content.

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523 TABLES

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Table 1 - Univariate analysis of Breast MRI features.

Variables	Category	SIGBIC (+)		P-value
		N (208)	%	
Capsular Contracture	(-)	0	0	<0.001²
	(+)	208	30.6%	
Intracapsular Rupture	(-)	207	100.0%	-
	(+)			
Extracapsular Rupture	(-)	208	30.8%	0.319 ²
	(+)	0	0.00%	
Water droplet	(-)	142	25.7%	<0.001¹
	(+)	66	51.97%	
Implant Rotation	(-)	198	31.2%	0.274 ¹
	(+)	10	22.2%	
Siliconoma Extracapsular	(-)	207	30.6%	1.000 ²
	(+)	1	25.0%	
Enlarged Intramammary Lymph Node (EILN)	(-)	179	28.3%	<0.001¹
	(+)	29	60.4%	
Pericapsular edema	(-)	179	27.9%	<0.001¹
	(+)	29	74.4%	
Intracapsular seroma	(-)	166	27.7%	<0.001¹
	(+)	42	51.2%	
Indication	Screening	11	9.8%	<0.001¹
	Diagnostic	197	35.2%	
Type of surgery	Aesthetic	156	28.5%	0.023¹
	Oncologic	52	39.1%	
Type of Implant	Silicone	207	31.2%	0.066 ²
	Saline	1	6.7%	
	Double	0	0.0%	
Location of Implant	Retroglandular	141	29.2%	0.290 ¹
	Retromuscular	66	33.7%	
Patient Age	Mean (S.E.)	43.8 (0.8)		0.026³
Time of surgery	Mean (S.E.)	7.7 (0.5)		0.878 ³

525 ¹Chi-Square test. ²Fisher's Exact test. ³Mann-Whitney test

526 SIGBIC: silicone induced granuloma of breast implant capsule

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Table 2. Multivariate analysis associating BMRI variables with and without gel bleeding

Variables		O.R.	C.I. (95%)	P-value
Water droplet	(-)	1.0	-	-
	(+)	2.8	[1.8; 4.4]	<0.001
Enlarged Intramammary Lymph Node	(-)	1.0	-	-
	(+)	3.1	[1.5; 6.1]	0.001
Pericapsular edema	(-)	1.0	-	-
	(+)	5.0	[2.3; 11.1]	<0.001
Intracapsular Seroma	(-)	1.0	-	-
	(+)	2.4	[1.4; 4.1]	0.001
Indication	Screening	1.0	-	-
	Diagnostic	5.1	[2.6; 10.0]	<0.001
Type of surgery	Aesthetic	1.0	-	-
	Oncologic	1.9	[1.2; 3.1]	0.006
Patient Age		1.0	[1.0; 1.0]	0.042

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O.R. odds ratio

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C.I. confidence interval

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548 **Table 3. Quality statistics of the multivariate model for GB diagnosis adopting water droplets,**
549 **enlarged intramammary lymph nodes, pericapsular edema and intracapsular seroma.**

SEN	SPE	PPV	NPV	Accuracy	AUC	Hosmer-Lemeshow	Pseudo-R ²
0.740	0.638	0.474	0.848	0.669	0.750	0.795	24.79%

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551 SEN: sensitivity

552 SPE: specificity

553 PPV: positive predictive value

554 NPV: negative predictive value

555 AUC: area under curve

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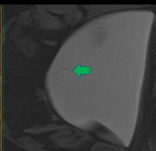
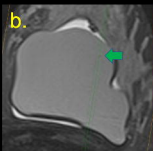
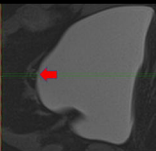
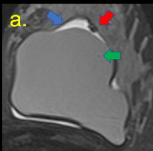
Table 4 - Univariate analysis of Breast MRI features.

Variables	Category	SIGBIC (-)		SIGBIC (+)		P-value
		N (473)	%	N (208)	%	
Capsular Contracture	(-)	123	26.1%	0	0	<0.001 ²
	(+)	349	73.9%	208	30.6%	
Intracapsular Rupture	(-)	439	68.6%	201	31.4%	0.094 ¹
	(+)	33	82.5%	7	17.5%	
Extracapsular Rupture	(-)	468	69.2%	208	30.8%	0.319 ²
	(+)	4	100.00%	0	0.00%	
Water droplet	(-)	411	74.3%	142	25.7%	<0.001 ¹
	(+)	61	48.0%	66	51.97%	
Implant Rotation	(-)	437	68.8%	198	31.2%	0.274 ¹
	(+)	35	77.8%	10	22.2%	
Siliconoma Extracapsular	(-)	469	69.4%	207	30.6%	1.000 ²
	(+)	3	75.0%	1	25.0%	
Enlarged Intramammary Lymph Node (EILN)	(-)	453	71.7%	179	28.3%	<0.001 ¹
	(+)	19	39.6%	29	60.4%	
Pericapsular edema	(-)	462	72.1%	179	27.9%	<0.001 ¹
	(+)	10	25.6%	29	74.4%	
Intracapsular seroma	(-)	432	72.2%	166	27.7%	<0.001 ¹
	(+)	40	48.8%	42	51.2%	
Indication	Screening	109	90.8%	11	9.8%	<0.001 ¹
	Diagnostic	363	64.8%	197	35.2%	
Type of surgery	Aesthetic	391	71.5%	156	28.5%	0.023 ¹
	Oncologic	81	60.9%	52	39.1%	
Type of Implant	Silicone	456	68.8%	207	31.2%	0.066 ²
	Saline	14	93.3%	1	6.7%	
	Double	2	100.0%	0	0.0%	
Location of Implant	Retroglandular	342	70.8%	141	29.2%	0.290 ¹
	Retromuscular	130	66.3%	66	33.7%	
Patient Age	Mean (S.E.)	41.6 (0.5)		43.8 (0.8)		0.026 ³
Time of surgery	Mean (S.E.)	7.3 (0.2)		7.7 (0.5)		0.878 ³

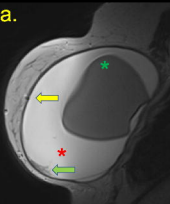
577 ¹Chi-Square test. ²Fisher's Exact test. ³Mann-Whitney test

578 SIGBIC: silicone induced granuloma of breast implant capsule

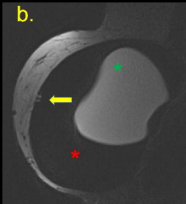
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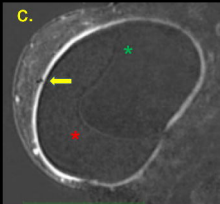
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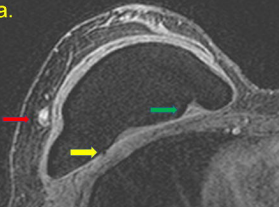
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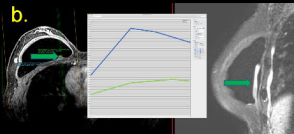
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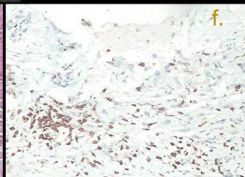
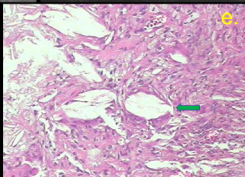
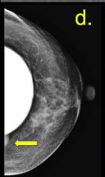
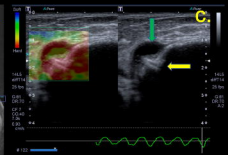
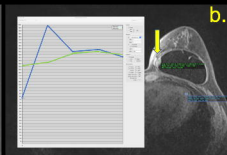
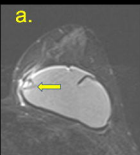


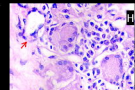
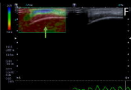
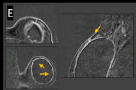
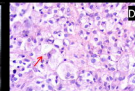
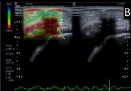
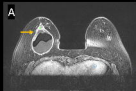
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